The Journal of Open Source Software

OptimalControl.jl: A Julia package for modeling and solving optimal control problems with ODEs

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DOI: 10.xxxxxx/draft

Software

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Editor: Open Journals

Reviewers:

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Submitted: 01 January 1970
Published: unpublished

Summary

OptimalControl.jl (Caillau, Cots, Gergaud, Martinon, & Sed, 2024) is a Julia (Bezanson et al., 2017) package for modeling and solving optimal control problems governed by ordinary differential equations (ODEs). As the core of the control-toolbox ecosystem, it provides a unified framework that supports both direct and indirect solution methods with applications spanning aerospace, medical imaging, epidemiology, and quantum control.

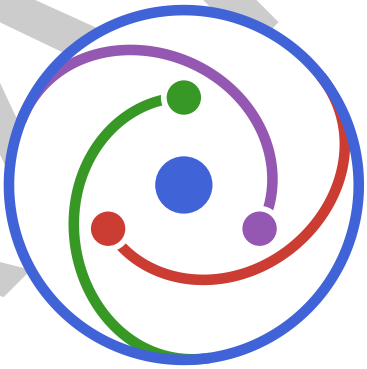

Figure 1: Logo of OptimalControl.jl.

The package features an expressive domain-specific language (DSL) built around the `@def` macro, enabling users to define control problems using notation that closely resembles standard mathematical formulations. Problems are solved through direct transcription, converting the continuous problem into a nonlinear program (NLP) using discretization schemes including Euler, trapezoidal, midpoint, and high-order Gauss-Legendre collocation. Alternatively, indirect shooting methods based on Pontryagin's Maximum Principle can be employed. The architecture relies on a modeler-solver separation that provides a modular and extensible foundation, enabling GPU execution (currently limited to NVIDIA hardware) with minimal user intervention alongside standard CPU solvers.

The ecosystem includes extensive tutorial resources, a benchmark problem collection (OptimalControlProblems.jl (Caillau, Cots, Gergaud, & Martinon, 2024) with formulations in OptimalControl DSL and JuMP (Lubin et al., 2023)), and performance comparison tools (CTBenchmarks.jl). Integration with Julia's ecosystem enables access to state-of-the-art tools: NLP solvers IPOPT (Wächter & Biegler, 2006) and MadNLP.jl (Shin et al., 2021, 2024), automatic differentiation and NLP modeling through JuliaSmoothOptimizers' ADNLPModels

(Migot et al., 2023), GPU acceleration via ExaModels.jl (Shin et al., 2023), numerical integration from SciML's DifferentialEquations.jl (Rackauckas & Nie, 2017), and visualization through Plots.jl (Christ et al., 2023).

Problem Class

The package currently targets single-phase problems governed by explicit ODEs, written in Bolza form,

$$g(t_0, x(t_0), t_f, x(t_f), v) + \int_{t_0}^{t_f} f^0(t, x(t), u(t), v)\,\mathrm{d}t \longrightarrow \min,$$

subject to the dynamics $\dot{x}(t) = f(t, x(t), u(t), v)$, where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ the control, and $v \in \mathbb{R}^q$ an optimisation variable, that is a finite-dimensional parameter optimised together with the trajectory. Mayer ($f^0 = 0$) and Lagrange ($g = 0$) costs are the corresponding special cases, and maximisation is also supported. Both autonomous and non-autonomous dynamics are handled, the dependences being detected by the parser. The time bounds t_0 and t_f may be fixed or be components of v, which covers free initial and / or final time, in particular minimum time problems. Admissible constraints are of five types — boundary, variable, control, state, and mixed (control and state), the last three being *path* constraints, imposed at all times — and may be equalities or one- or two-sided inequalities, either box constraints on state, control and variable (for which dedicated bounds are passed to the NLP solver) or nonlinear. The variable may appear in any of them. The current implementation is restricted to *single-phase* problems with *explicit* ODE dynamics: multiphase problems, differential-algebraic equations, and hybrid or nonsmooth systems are not supported. Optimization of PDE or stochastic systems is out of scope as well (see the design trade-offs below).

Statement of Need

OptimalControl.jl is, to our knowledge, the only Julia package that unifies both direct and indirect methods for optimal control within a single, coherent framework.

This unified approach addresses a gap in the current optimal control software landscape. Existing tools are fragmented across programming languages and paradigms. Legacy packages such as HamPath (Caillau et al., 2012), NutoPy (Caillau et al., 2022), and COTCOT (Bonnard et al., 2007) implement sophisticated indirect methods but rely on Fortran implementations with MATLAB or Python interfaces, leading to complex and less extensible workflows. Proprietary solvers like GPOPS-II (Patterson & Rao, 2014) limit transparency and reproducibility. Open-source tools such as BOCOP (Bonnans et al., 2017), ACADO (Houska et al., 2011), and acados (Verschueren et al., 2021) provide valuable direct method implementations but are not natively integrated into a modern, high-level scientific computing ecosystem. Python-based tools such as Dymos (Falck et al., 2021), Pyomo.DAE (Nicholson et al., 2018), do-mpc (Fiedler et al., 2023), and GEKKO (Beal et al., 2018) offer complementary capabilities but are similarly limited to direct transcription approaches within the Python ecosystem.

Within Julia, most existing packages target specific domains: RobustAndOptimalControl.jl for linear systems, QuantumControl.jl (Machnes et al., 2025) and Piccolo.jl for quantum optimal control, DirectTrajectoryOptimization.jl for trajectory problems, LinearMPC.jl (Arnström et al., 2022) and ModelPredictiveControl.jl (Gagnon et al., 2024) for model predictive control, and InfiniteOpt.jl (Pulsipher et al., 2022) for infinite-dimensional optimization. In contrast to InfiniteOpt.jl, which is designed as an extension of JuMP.jl, our package provides a modeler-agnostic approach, accepting general Julia code and leveraging various optimization solvers. Moreover, contributing to general-purpose NLP modeling frameworks like JuMP or CasADi

(Andersson et al., 2019) would not address the specific needs of optimal control: these tools lack native support for shooting methods based on Pontryagin's Maximum Principle, which heavily rely on differential geometric primitives.

OptimalControl.jl fills this gap by providing a DSL that matches mathematical notation, multiple discretization schemes and shooting methods, with planned support for homotopy continuation methods. The modeler-solver separation enables complementary use with InfiniteOpt.jl (JuMP-based) through alternative NLP modeling backends (currently ADNLPModels, ExaModels) and solvers. GPU acceleration and an ecosystem with domain-specific applications, tutorials, and benchmarking tools complete the offering. Target users include researchers and engineers working in optimal control, control theorists developing new algorithms, and students learning optimal control through interactive tutorials.

State of the Field

OptimalControl.jl requires Julia version 1.10 or later and is registered in the Julia General registry, enabling straightforward installation via `Pkg.add("OptimalControl")`. We compare it below with existing software.

- **Legacy tools (COTCOT, HamPath, NutoPy)**: These Fortran packages excel at indirect methods and homotopy continuation but require multi-language setup (Fortran plus MATLAB / Python). OptimalControl.jl provides both direct and indirect methods in pure Julia with straightforward installation via package manager.
- **Direct-method tools (BOCOP, ACADO, GPOPS-II, acados, nosnoc)**: Strong direct-method implementations: GPOPS-II delivers mature methods with MATLAB and C++ implementations but is proprietary; acados targets real-time MPC on embedded systems; nosnoc (Nurkanović & Diehl, 2022) specializes in nonsmooth optimal control; CasADi, used as symbolic backend by several of these tools, is a general NLP modeler rather than an optimal control solver. OptimalControl.jl offers an open-source alternative with expressive DSL, native Julia ecosystem integration, GPU support, and unified direct and indirect approaches.
- **Julia packages**: RobustAndOptimalControl.jl targets linear systems; QuantumControl.jl, Piccolo.jl and DirectTrajectoryOptimization.jl serve specific domains; LinearMPC.jl and ModelPredictiveControl.jl focus on model predictive control. InfiniteOpt.jl addresses a very rich range of problems, including optimization on PDEs or with chance constraints, focusing on direct transcription methods. OptimalControl.jl supports both CPU and GPU execution (currently NVIDIA-only, via ExaModels.jl + MadNLP.jl), and adds tools to do shooting in a unified framework plus systematic benchmarking through OptimalControlProblems.jl and CTBenchmarks.jl.

Illustrative Example

A self-contained illustrative example is provided as a companion repository (Caillau et al., 2026), archived on Zenodo. It combines direct and indirect solution approaches for a constrained energy minimization problem: a direct method on a coarse grid identifies the three-arc structure (unconstrained–constrained–unconstrained) and initializes a shooting method based on Pontryagin's Maximum Principle, which then converges to arbitrary precision.

Software Design

The package architecture balances expressiveness, performance, and extensibility through modular design. The core is organized across internal packages within the control-toolbox organization, each hosted as an independent Julia package and registered in the General

registry: CTBase.jl defines the core types (optimal control models, solutions, initial guesses) and their accessors; CTParser.jl implements the `@def` DSL macro; CTDirect.jl handles direct transcription and NLP interfacing; and CTFlows.jl provides Hamiltonian flows and shooting for indirect methods. OptimalControl.jl re-exports these packages as a unified entry point. Contributors interested in a specific functionality can work directly on the relevant sub-package, each of which has its own documentation, tests, and continuous integration. The delegation of specific responsibilities is as follows:

- **DSL parsing**: MLStyle.jl (Thautwarm & contributors, 2023) enables pattern matching on abstract syntax trees generated from `@def` macro invocations. This design choice separates problem specification from solution methods, and allows a syntax as close as possible to the mathematical description of the problem.
- **Problem models and solutions**: Structured types represent optimal control models, solutions, and initial guesses. Each type provides textual visualization, accessor methods for introspection, and visual plotting capabilities for solutions. Extensible abstractions enable addition of new problem classes without modifying core code.
- **Discretization**: Direct transcription converts continuous problems into NLPs. Supporting multiple discretization schemes (Euler, midpoint, trapezoidal, Gauss-Legendre) required careful abstraction to share transcription logic while allowing scheme-specific implementations.
- **Modelers and solvers**: We chose a clear modeler-solver separation: discretization produces ADNLPModels (able to deal with arbitrary user-defined functions) or ExaModels instances compatible with multiple NLP solvers (IPOPT via NLPModelsIpopt.jl (Orban et al., 2023), Knitro (Byrd et al., 2006), MadNLP, and Uno (Vanaret & Leyffer, 2026)). These NLP solvers rely on linear solvers such as MUMPS.jl (Amestoy et al., 2001, 2019; Montoison et al., 2024) for CPU computations or CUDSS.jl (NVIDIA Corporation, 2025) for GPU acceleration with MadNLP. Automatic differentiation via ForwardDiff.jl (Revels et al., 2016) and DifferentiationInterface.jl (Dalle & Hill, 2026) avoids manual derivative coding.
- **Indirect methods**: Hamiltonian flows integrate with DifferentialEquations.jl to access adaptive stepping, event handling, and multiple ODE solvers without reimplementation. Shooting methods rely on NonlinearSolve.jl (Pal et al., 2025) or MINPACK.jl (Moré et al., 1980) for root-finding. Future extensions will incorporate homotopy continuation methods leveraging bifurcation analysis tools like BifurcationKit.jl (Veltz, 2020). Currently, indirect methods are limited to Hamiltonian flows, where the feedback control law depends on both state and costate; support for non-Hamiltonian flows (open-loop or state-feedback control laws) is under active development.
- **Testing and quality assurance**: The multi-repository structure is matched by a layered testing strategy. Each sub-package is tested independently, combining unit tests, integration tests, and code-quality checks, while OptimalControl.jl adds strong end-to-end integration tests that solve complete optimal control problems by both direct and indirect methods. Continuous integration runs across Linux, macOS, and Windows, on both CPU and GPU (through a self-hosted CUDA runner), using reusable GitHub Actions workflows centralized in CTActions and shared across the ecosystem. Code coverage is tracked on Codecov, downstream packages and applications are guarded against regressions through dedicated breakage tests triggered on pull requests, and beta versions are distributed during development via a local registry, ct-registry. Regression testing on complex optimal control instances is increasingly supported by OptimalControlProblems.jl, which serves as a curated problem bank.

Key design trade-offs:

1. **DSL vs. programmatic API**: We prioritized a mathematical DSL to reduce cognitive load for domain experts, accepting increased parsing complexity handled internally. Currently

we do not address optimization of PDE or stochastic systems.

2. **GPU acceleration strategy**: The modeler-solver separation enables a straightforward CPU-to-GPU transition. Users select the ExaModels modeler with MadNLP and the CUDSS linear solver for GPU execution, or simply append the `:gpu` token to the `solve` call. This modular approach minimizes maintenance burden while enabling GPU performance without reimplementing transcription logic. GPU support is currently limited to NVIDIA hardware and to the ExaModels + MadNLP modeler-solver combination.

3. **Method coverage**: Supporting both direct and indirect approaches increases code complexity but serves distinct user needs: direct methods for constrained problems with many variables, indirect methods for theoretical analysis and smaller problems requiring high accuracy. Both approaches rely on iterative solvers and may or may not converge, *e.g.*, depending on the initial guess. In the case of optimization solvers, the full output status of the solver is returned allowing *a posteriori* analysis.

Research Impact Statement

OptimalControl.jl has been applied in published research across multiple domains. Independent external adoption demonstrates the package's broader impact: Ferede et al. (2025) used it for drone racing trajectory optimization, Caio & Silva (2025) for epidemiological models using indirect methods, Morsky (2025) for vaccination strategies under social norms, Opmeer (2025) for optimal harvesting of age-structured populations, and Evangelakos et al. (2025) for fast charging protocols for quantum batteries.

Research by close collaborators further demonstrates the package's versatility across aerospace (Herasimenka et al. (2026)), epidemiology (Bliman et al. (2026)), quantum control (Beschastnyi et al. (2025)), microbial growth control (Innerarity Imizcoz et al. (2025)), and theoretical optimal control (Bouali et al. (2025); Lutz & Privat (2025); Bayen et al. (2026); Bonnard et al. (2026)).

The control-toolbox organization hosts domain-specific application packages built on OptimalControl.jl: medical imaging optimization (MagneticResonanceImaging.jl), gene regulatory networks (PWLdynamics.jl), spacecraft orbital transfers (Kepler.jl), epidemiological modeling (SIRcontrol.jl), and variational calculus (CalculusOfVariations.jl).

The package also serves educational purposes through Tutorials.jl, which covers topics from linear-quadratic regulators to Model Predictive Control. These resources are used in academic courses and workshops; the package was notably presented at JuliaCon 2023 (Caillau et al., 2023). OptimalControlProblems.jl provides standardized test problems formulated in both OptimalControl DSL and JuMP, enabling systematic performance comparisons through CTBenchmarks.jl.

Development involves international collaborations with CNES (French space agency), Thales Alenia Space, Inria, and CNRS. The package development began in September 2022, with the first public release in February 2023. As of March 2026, OptimalControl.jl has published more than 40 releases with contributions from multiple developers across the control-toolbox ecosystem, demonstrating sustained community engagement.

AI Usage Disclosure

The core software implementation, algorithms, and architectural design of OptimalControl.jl were developed by the authors. Generative AI tools—specifically Claude Sonnet (version 4.6) and Claude Opus (version 4.8) by Anthropic—have been used to assist with code refactoring and generation of unit and integration tests, always under human review. This paper was drafted by the authors and subsequently revised with AI assistance for restructuring according

to JOSS format requirements and language editing. All technical content, examples, and claims reflect the authors' work and judgment.

Acknowledgements

We thank the control-toolbox community and the broader Julia ecosystem for contributions that shaped the package design and educational resources. Development has been supported by partnerships with CNES, Thales Alenia Space, Inria, and CNRS. The software is distributed under the MIT license (see LICENSE). We welcome community contributions following the guidelines in CONTRIBUTING.md.

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